Results and Analysis:

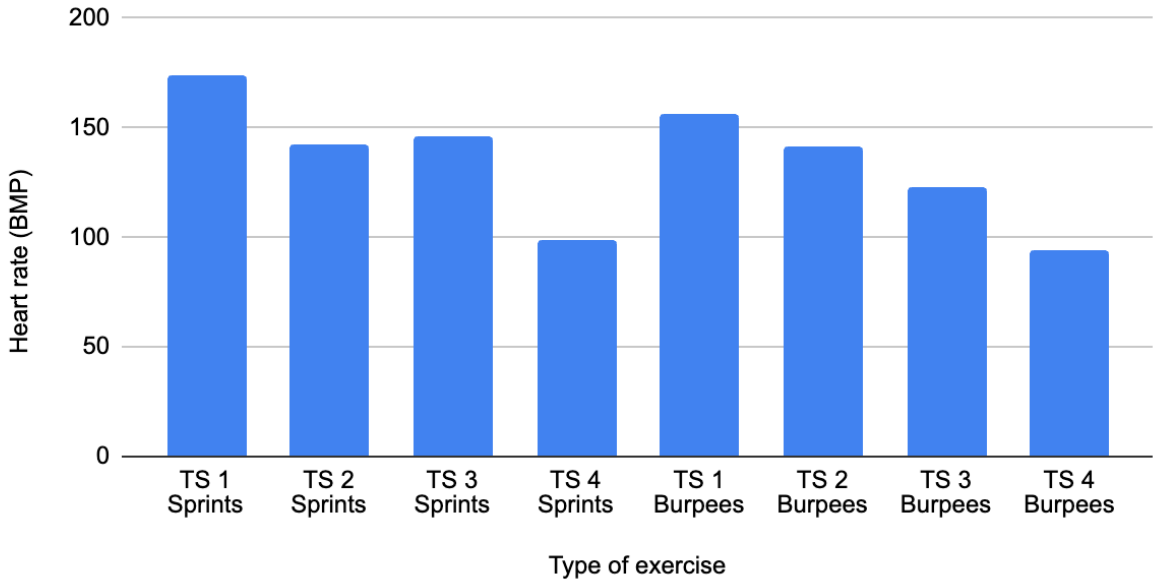
- Must contain a complete set of quantitative data (three trials of the experiment and average data) and average data in a data table with table number, descriptive title, and units and values in the correct places
- Must contain a graph of the average data with the x and y-axis labelled with units, a figure number and descriptive title (you must decide the appropriate type of graph to use for your data)
- Can contain a table of qualitative data (if necessary) with Table number and descriptive title
- Must contain a summary of the trends shown in the data collected

Insert data table(s) in the box below:

	TS 1 Sprints Jacklyn	**TS 2 Sprints Harry**	**TS 3 Sprinting Jonah**	**TS 4 Sprinting Dina**
Heart rate (BPM), two sprints (23.5 meters)	179BPM Trial 1 166BPM Trial 2 176BPM Trial 3	142BPM Trial 1 140BPM Trial 2 145BPM Trial 3	152BPM Trial 1 142BPM Trial 2 145BPM Trial 3	90BPM Trial 1 97BPM Trial 2 108BPM Trial 3
Average	**173.67BPM**	**142.33BPM**	**146.33BPM**	**98.33BPM**
Heart rate (BPM) of 20 no-push-up burpees	156BPM Trial 1 160BPM Trial 2 153BPM Trial 3	141BPM Trial 1 143BPM Trial 2 139BPM Trial 3	132BPM Trial 1 124BPM Trial 2 112BPM Trial 3	90BPM Trial 1 95BPM Trial 2 97BPM Trial 3
Average	**156.33BPM**	**141BPM**	**122.67BPM**	**94BPM**

Insert graph(s) in the box below:

Figure 1. The average Heart Rate (PM) of Sprinting and Burpees



Summary of trends in the data:

- Accurately interpret data and **describe** results using correct scientific reasoning

The results show that the heart rate was higher after two sprints compared to 20 burpees. The average heart rate for sprints was about 140.67 BPM while for burpees it was 128.50 BPM. That means sprinting made peoples heart rate go up more, probably because it uses more energy and faster muscles. The sprints were shorter but more intense, which causes the heart to pump blood faster to get oxygen to the muscles (NIDDK). This also shows that different types of exercises can effect how fast your heart beats, even if they both feel hard (American Heart Association).

Comparison, Analysis, and Conclusion:

- Accepts or rejects the hypothesis while re-stating the hypothesis
- **Discuss**, by comparing the data collected on all variables, why the hypothesis was proven correct or incorrect, and **draw a conclusion** based on the data

Hypothesis:
If two sprints were performed, then it would have a lower heart rate, because burpees involve more varied movements.
Conclusion:
The hypothesis was incorrect. The experiment showed that the heart rate was higher after two sprints than after twenty burpees. This could be because sprinting is a short,

high-intensity exercise that makes your body use energy faster (NIDDK). Burpees, even though they have more movements, are spread out and not as fast, so the heart doesn't have to work as hard at once. That means that sprinting actually caused more strain on the heart, not less (American Heart Association).

Evaluation of Method:

- **State** two sources of experimental error
- **Discuss** the problem caused by this error with respect to the data collection
- **Discuss** ways to fix each error for future experiments

Error 1:

Source of Error:	**The Problem(s) caused by this error:**	**Way to fix the error**
Not everyone did burpees or ran with the same effort	If someone didn't run at full speed or breathe faster/slower, the outcome would be different.	Make sure everyone does the exercise at full effort

Error 2:

Source of Error:	**The Problem(s) caused by this error:**	**Way to fix the error**
Heart rate wasn't always measured right away after finishing the exercise.	Heart rate can drop really quick after you stop moving, so some results might be a little lower than they really were.	Use a smartwatch or heart rate tracker that measures it instantly after finishing. That way the numbers will be more accurate and you'll get the true result.

MLA Works Cited:

- Must contain at least two relevant sources
- Must be alphabetized by last name of the author of first letter of article title; whichever you have
- Must follow MLA format

"Exercise Intensity and Heart Rate." *National Institute of Diabetes and Digestive and Kidney Diseases*, 2022, niddk.nih.gov/health-information/weight-management/tips-get-active/exercise-intensity

.“Measuring Heart Rate: Exercise and Your Pulse.” *American Heart Association*, 2023, heart.org/en/healthy-living/fitness/fitness-basics/measuring-heart-rate.